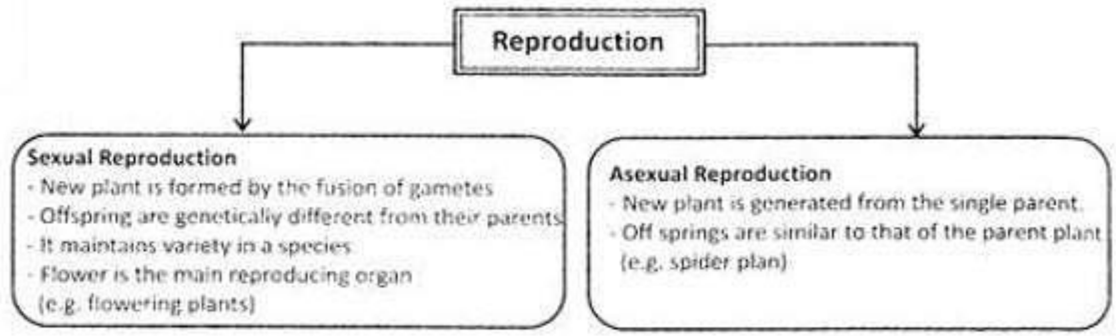
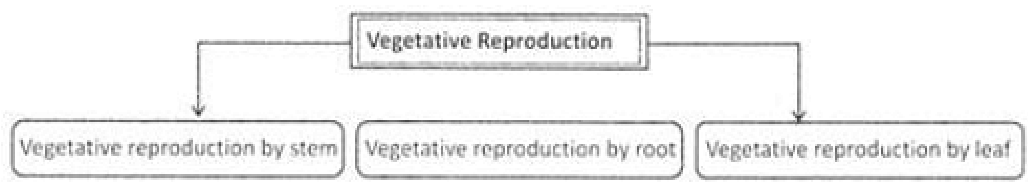
Plant Life (Part - 4)

Reproduction:
Reproduction is the process of producing offspring of its own kind. For example an apple tree can produce another apple tree only, it can't produce mango tree.



Vegetative reproduction is a type of asexual reproduction in which new plant is generated from the vegetative part of the plants i.e. stems, roots, leaves etc.



Vegetative reproduction by Stem
Runnerss

Stems, which run horizontally parallel to the ground. New roots emerge from the nodes, thus gives rise to new plants e.g. lawn grass, strawberries etc.

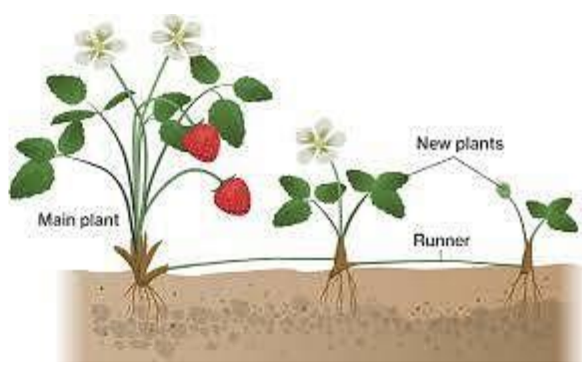
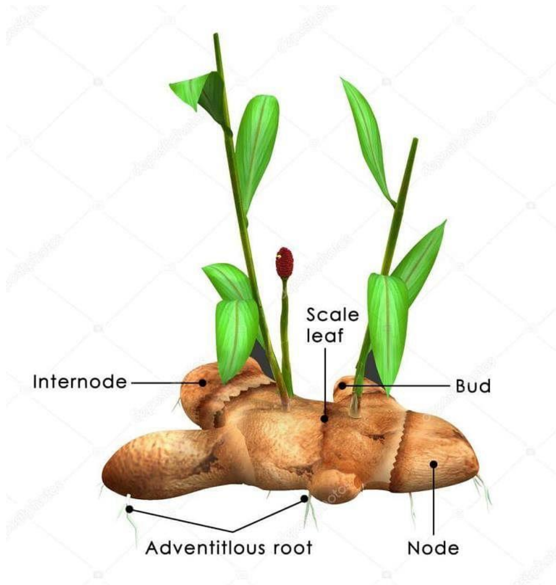


Fig: Reproduction in Strawberry

The reproduction is very similar to runners but have shorter internodes, e.g. mints, chrysanthemum.



Reproduction in ginger by rhizome

It has underground nodes and internodes. Axillary buds are present on the nodes, which develop into new plants on getting favorable conditions, e.g. ginger.

Tuber:

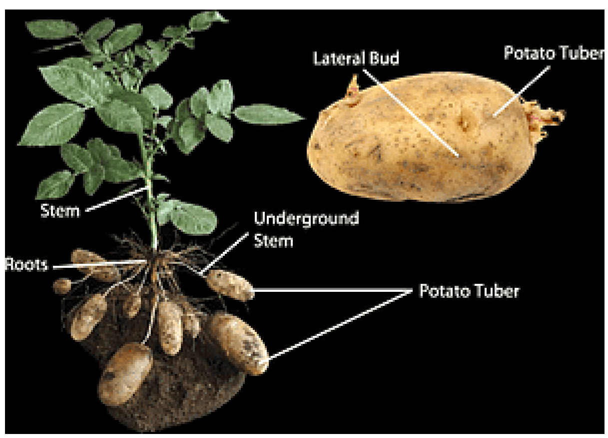
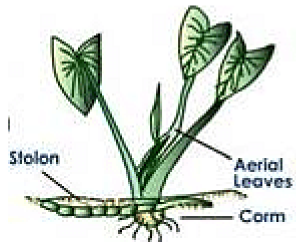


Fig: Vegetative reproduction in Potato

Potato is a tuber that has bud, which is capable of producing new plant.

Corn:

Corm is a condensed form of rhizome, which is formed vertically in the ground. A single corm develops new plant e.g. colocasia

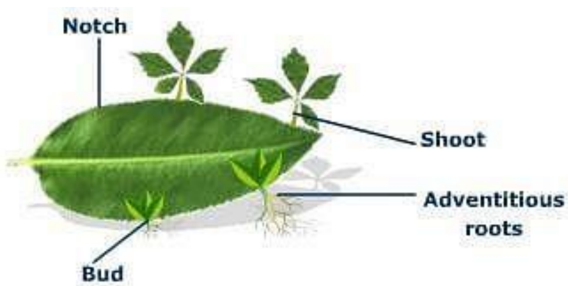


Reproduction in colocasia

Bulbs:

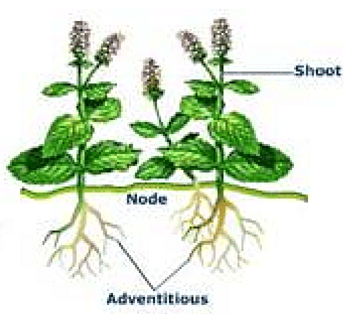
A type of plants which has bulbs for the storage of food for helping them in non favorable condition. The bud in the plants grows into new plants, e.g. onion, lily, etc,

Vegetative Reproduction by Leaf

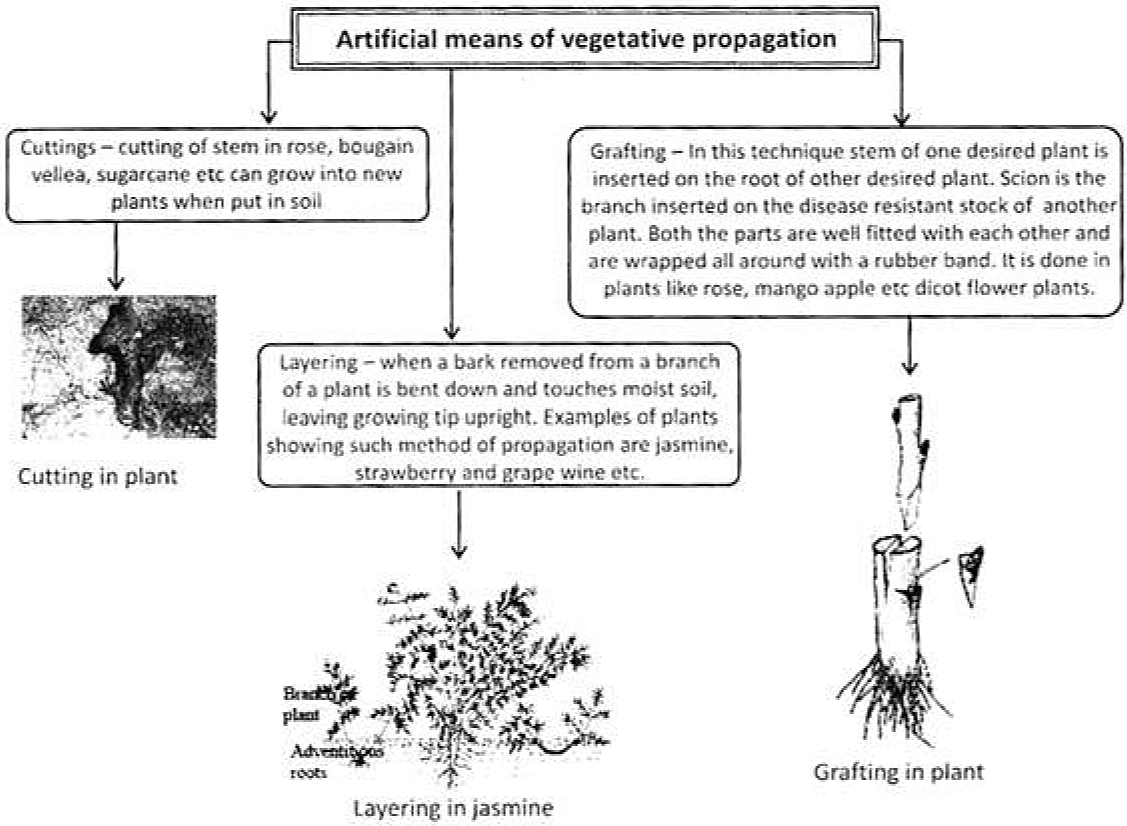


Adventitious Buds:
Buds present on the margin of leaves, which develops into new plants on getting the favorable condition.

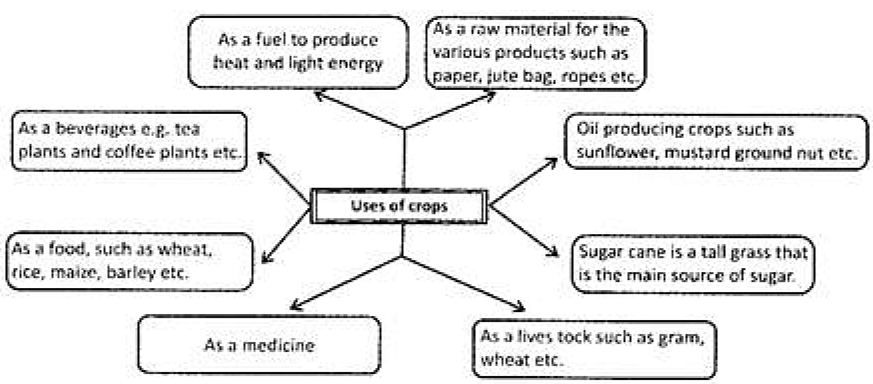
Vegetative Reproduction by Roots:



Vegetative reproduction in a plant is the asexual reproduction. Tuberous roots of sweet potato and dahlia when get detached from the plant grow into new plant.



Uses of crops:



Medicinal plants:

Bael or wood apple good for digestion process

Ashwagandha of winter chery Good for treating arthritis and diabetes

Aloe Vera-good for treating cuts and burns. skin problems etc.

Margosa-good for Treating bacteria, virul, fungal etc. infection, good for dental care also

Commonly Asked questions:

1. Which one of the following is an oil yielding crop?

(a) Cotton
(b) Castor
(c) Rapeseed
(d) Soya bean
(e) All of these

Answer: (e)

Explanation

Oil can be extracted from the following seed: Castor, copra, cotton, groundnut, palm, mustard, rapeseed, sesame, soybean and sunflower.

Therefore, option E is correct and rest of the options is incorrect.

Find the incorrect statement about the crops.

(a) Rice is the main food crop in Asia
(b) Tea is made from the leaf tips that are harvested, dried and then crushed
(C) The main source of the sugar is the sweet stems of sugar cane
(d) Chocolate is made from the cacao tree
(e) Coffee is obtained from the leaves of the coffee plant

Answer: (e)
Explanation
Berries of coffee plant is used to make coffee. Ripe berries are harvested, dried and then processed to make coffee. Therefore, option (E) is correct and rest of the options is incorrect,

You Must Know:

1. Venus fly trap, pitcher plant and bladder worth etc. derive their nitrogen requirement from the insects.
2. Bamboo is not a tree but a grass
3. The flower of Raffles is the largest known flower
4. Edible mushroom grows completely within 7 days.
5. Amount of carbon dioxide produced by plants are less than that of oxygen produced by them

Plants:
Plants are very useful to us in many ways. Being a living thing, like all other living things, plants also reproduce to maintain the existence of their species. Plants reproduce in a number of ways:
(i) From their body parts
(ii) From spores
(iii) From seeds
Reproduction through Body Parts:
Some new plants grow from the parts of the mother plants. They grow from roots, stem and leaf. Such type of reproduction is known as vegetative propagation.
Root:
A sweet potato is a swollen root of the plant. The new plant of sweet potato can be grown from the roots of the original plant.

Fig: Sweet Potato

Stem:
Some plants can reproduce by burying a part of stem in the soil. From stem cutting, new shoots grow from buds. For example,

Fig; Rose Plant

Underground Stem:
Underground stems like potato and ginger have buds on them from where new shoots will grow on planting them in the soil.

Fig: Potato

Similarly, some plants like onions and lilies grow from their bulb shaped stems.

Leaves:
In Bryophyllum, new plant grows from their leaves

Fig: Begonia plant

Spores:
The plants like ferns, mushrooms or mosses, which do not have flowers, produce spores which can be grown into a new plant.

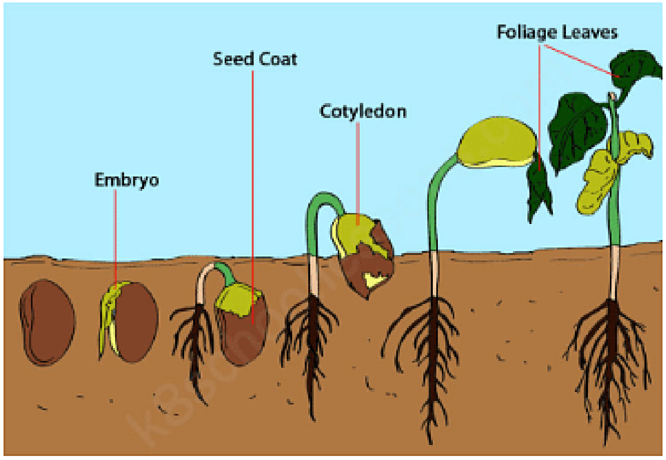
Fig: Mushrooms

Seeds:
The fruit bearing plants have seeds inside fruit. When these seeds fall on the soil, new plants grow from them. For example, Mango, rice, wheat and tomato, etc.

Fig: Seeds of an apple

Parts of Seed:
Seed has an upper covering which is called seed coat. Inside the seed coat, there may be one or two seed leaves called cotyledons. Between them, baby plant grows called seedling. Seedling has radicle growing downwards that develops into root and plumule growing upwards towards the sunlight that develops into shoot. Food for the baby plant is stored in the cotyledons. Number of cotyledons also varies from plant to plant. Maize, wheat and rice have only one cotyledon and are monocot plants. Pea and beans have two cotyledons and are dicot plants.
The process by which baby plant grows from a seed is called germination.
A seed needs water, sunlight and air. A seedling grows into a plant when it gets

sufficient water and food from soil, sunlight and air.



Germination of seeds Dispersal of Seeds:
Plants have to disperse their seeds with the help of nature so that all seeds would not fall on one place causing lack of sunlight, water and space to grow. Seeds are dispersed by wind, water, animals and explosion of fruits. These all are agents of dispersal.
Wind:
The seeds which are light and have wings or hairs on them are easily, carried by wind. Seeds of cotton and madar have hairs.

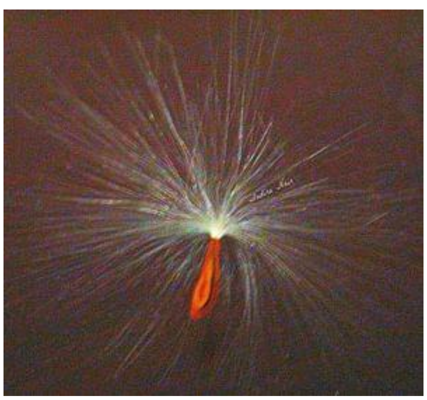


Fig: Madar Seed

Fig: Cotton Seed

Water:
Seeds are carried to different places by flowing water.
Coconut is very light in weight as it is hallow from inside so that it does not sink when it falls into water. It floats for several months in sea before reaching land.
Animals:
Seeds of mango, apple, etc. are thrown by human beings and animals after eating.
Some seeds have stiff hair which cling to bodies of animals or feathers of birds or our clothes and carried away from one place to another.

Explosion:
The fruits of some plants, like pea on drying explode suddenly. This explosion causes the seeds to scatter away from the mother plant. For example, pea plant and bean plant.

Fig: Bean Plant

Fig: Pea Plant

Food from Plants:
The growth of different crops depends on different seasons, climate and soil. The crops which grow in winter from november to april are called rabi crops. Examples of rabi crops are wheat and gram.

Fig: Wheat

Fig: White Gram

The crops which grow in summer from June to October are called kharif crops. Rice, maize and bajra are examples of kharif crops.

Fig: Rice (A Kharif Crop)

Fig: Corn (A Kharif Crop)

Protection of Crops:

- Farmers make fences around the fields to keep them safe from animals.
- Pests like aphids and caterpillar can damage the crops. So, it is necessary to kill them by using pesticides. For example, atrazine, alachlor.
- When the grain has ripened, it has to be harvested and stored. Farmers keep them safe from moisture and animals like birds, rats and insects.
- Fruits and vegetables are stored in cold storage. Dry grains and pulses are stored in granaries.